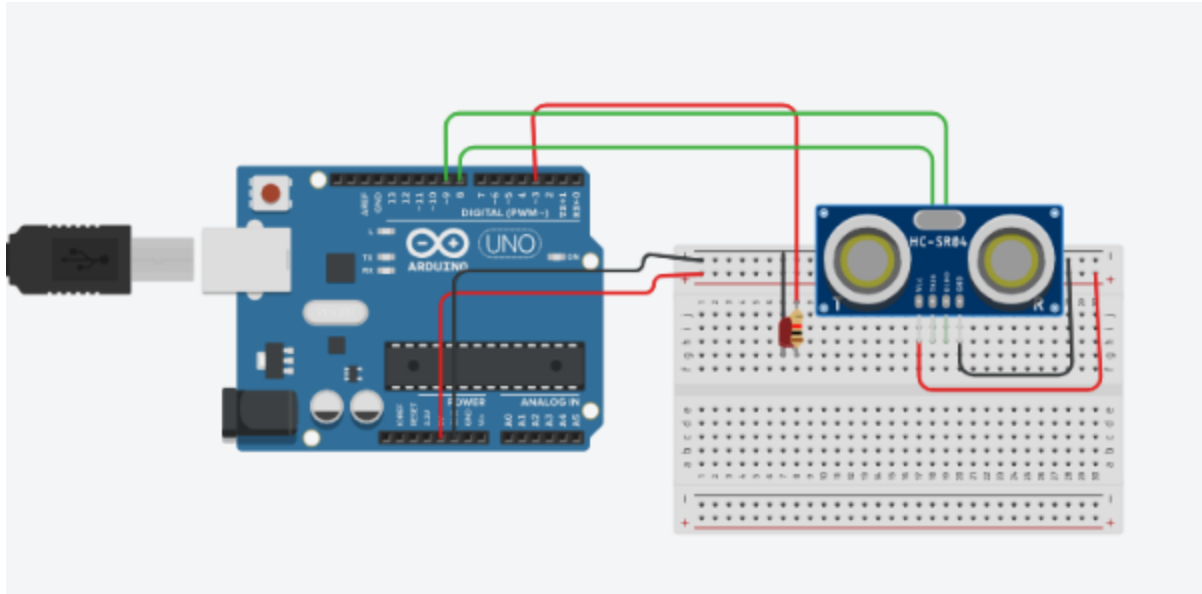
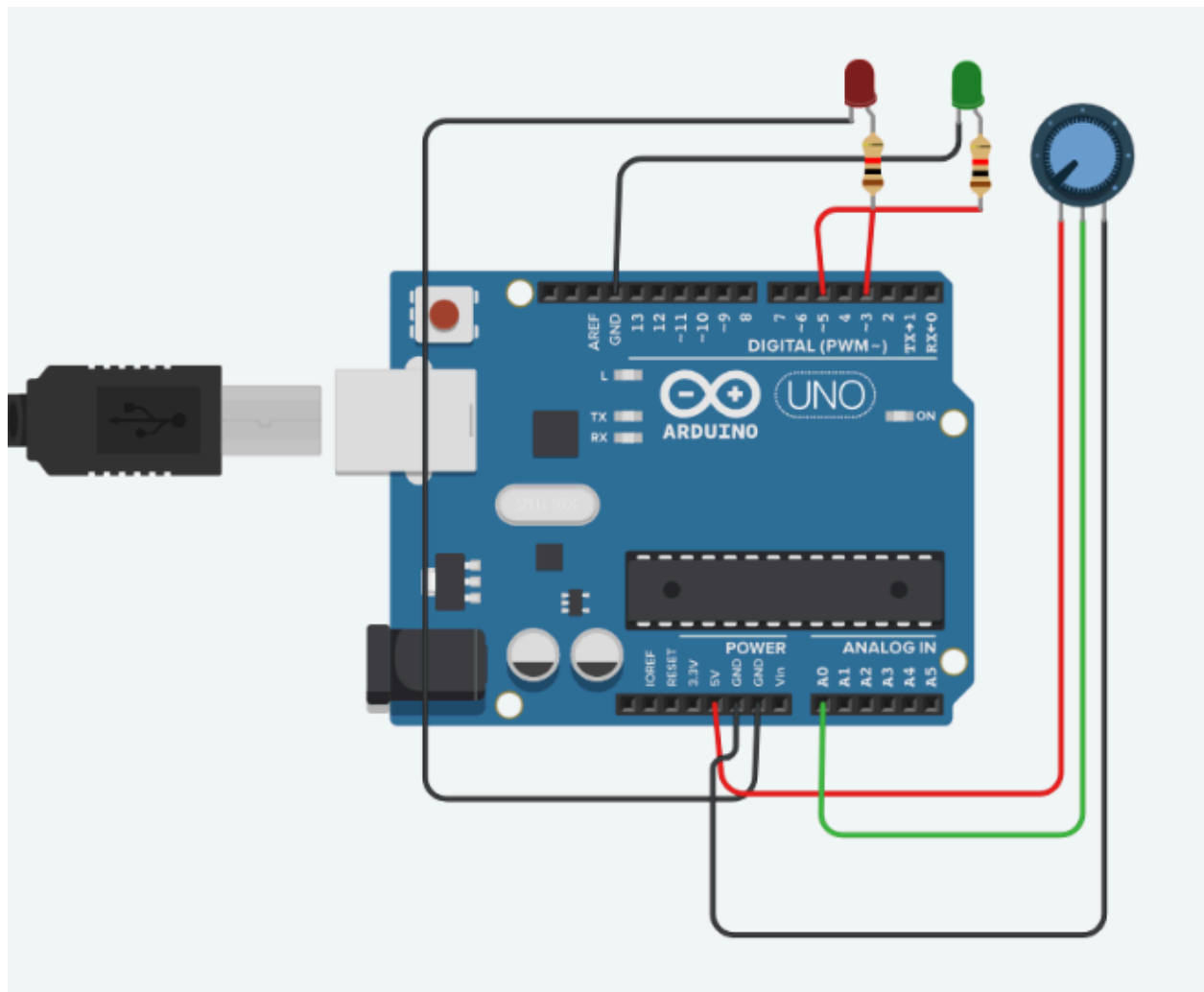




```
// C++ code
//
int red=3;
int trig=8;
int echo=9;int duration;
int distance;
float x;
void setup()
{
  pinMode(red, OUTPUT);
  pinMode(trig, OUTPUT);
  pinMode(echo, INPUT);
  Serial.begin(9600);
}
float mymap(float val,float ogmin,float ogmax,float usmin,float usmax)
{
  return (val-ogmin)*(usmax-usmin)/(ogmax-ogmin)+usmin;
}

void loop()
{
  digitalWrite(trig,LOW);
  delayMicroseconds(2);
  digitalWrite(trig,HIGH);
  delayMicroseconds(10);
  digitalWrite(trig,LOW);
  duration=pulseIn(echo,HIGH);
  distance=(duration*0.034/2);
```

```
Serial.println(distance);  
delay(500);  
if(distance<=20||distance>200)  
{  
    digitalWrite(red,LOW);  
  
}  
else  
{  
    x =mymap(red,20,200,0,255);  
    analogWrite(red,x);  
  
}  
  
}
```



```
// C++ code
//
int red=3;
int green=5;
int pw;
int x,y;
void setup()
{
  pinMode(red, OUTPUT);
  pinMode(green, OUTPUT);

  Serial.begin(9600);
}

void loop()
{
```

```
pw=analogRead(A0);
```

```
  x= map(pw,0,1023,0,255);  
  analogWrite(red,x);
```

```
  if(pw>0)
```

```
  {
```

```
    y= map(pw,0,1023,255,0);  
    analogWrite(green,y);
```

```
  }
```

```
}
```